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H2

Question 1 | Correct | Marked out of 1.00

Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  int compare(const void *a, const void *b) {
4      return (*(int*)a - *(int*)b);
5  }
6  int main() {
7      int n;
8      if (scanf("%d", &n) != 1) return 0;
9      int *arr = (int*)malloc(n * sizeof(int));
10     for (int i = 0; i < n; i++) {
11         scanf("%d", &arr[i]);
12     }
13     qsort(arr, n, sizeof(int), compare);
14     long long max_sum = 0;
15     for (int i = 0; i < n; i++) {
16         max_sum += (long long)arr[i] * i;
17     }
18     printf("%lld\n", max_sum);
19     free(arr);
20     return 0;
21 }
```

Check

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓

	Input	Expected	Got	
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

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